

Technology Arts Sciences TH Köln

M.Eng Automation and IT CNC Machine Case Study Report

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1 Introduction

This project documents the engineering workflow, configuration, and virtual commissioning of a SINUMERIK ONE CNC system in Siemens Create MyVirtual Machine (CMVM). The work follows the “Work steps for configuring and commissioning SINUMERIK ONE” commissioning manual as a reference workflow, but the scope of this case study is limited to the virtual commissioning phase only. No transfer to or commissioning on real SINUMERIK ONE hardware was performed because the control cabinet and operator panel hardware were not available during the laboratory implementation. The engineering process is structured into clearly defined phases: preparation of the engineering environment and toolchain (TIA Portal, SINUMERIK Toolbox, Create MyVirtual Machine), engineering of the PLC, NC, HMI on the virtual NCU. This component-oriented procedure reduces commissioning risk and effort by allowing most configuration and logic testing to be carried out on the digital twin. [2]

2 Components detail

2.1 Siemens NCU 1760 – Numerical Control Unit

[1]

2.1.1 Introduction

The Siemens NCU 1760 is a high-end Numerical Control Unit of the SINUMERIK ONE family, designed for complex, highly dynamic CNC machines with large axis counts and demanding motion requirements. It combines an advanced CNC kernel, a powerful integrated safety PLC, and high-performance drive communication into one compact control platform that is typically used in multi-channel machining centers, transfer lines, grinding centers, and high-speed milling applications.

As the central control element of the CNC system, the NCU 1760 executes all real-time tasks for machining, including multi-axis interpolation, kinematic transformation, spindle and feed control, tool-center-point management, and machine safety supervision. Its increased computing performance compared to lower NCU variants allows shorter block processing times, higher interpolation rates, and therefore better surface quality and reduced cycle times in demanding applications.

NOTE: We are using NCU 1760 specifically for our case study implementation as there is a firmware mismatch between the TIA V20 and CMVM(Create my virtual machine). The NCU1740 in TIA support firmware 6.22 and 6.25 among other firmware versions of the NCU1740 but the CMVM only has 6.22HF5 for NCU1740. To test the simulation, we are using NCU1760 firmware version 6.25.

2.1.2 Functional Architecture of the NCU 1760

The NCU 1760 integrates three main subsystems in a single module, analogous to the NCU 1740 but scaled for higher performance and axis counts:

CNC Core (Numerical Control Kernel)

- Executes CNC part programs in real time and manages path planning, axis interpolation, and block look-ahead.
- Supports multi-channel machining with several simultaneously active machining channels, each with its own axis set and spindle configuration.
- Provides advanced motion functions such as 5-axis transformation (TRAORI), spline interpolation, high-speed settings (e.g. COMPCURV/COMPCAD), and optional specific technology cycles (milling, turning, grinding).

Integrated PLC (SIMATIC S7-1500F)

- Safety-capable PLC (F-CPU) integrated on the same board, responsible for machine sequence logic, interlocks, and peripheral control.
- Programmed and diagnosed in TIA Portal with the usual S7-1500 programming languages (LAD, FBD, STL, SCL, etc.), including safety program sections.
- Provides a unified engineering concept: CNC, PLC, safety, and HMI are configured in a consistent project environment, simplifying commissioning and maintenance.

Communication and I/O Management Layer

- Handles real-time communication with SINAMICS drives and distributed I/O via PROFINET and DRIVE-CLiQ.
- Provides Ethernet-based interfaces for HMI, service access, and higher-level IT/MES integration.
- Manages onboard and distributed digital/analog I/O images, synchronizing process data exchange with the CNC and PLC cycles.

2.1.3 System Capabilities and Performance

Multi-Axis Machining Support

- Designed for machines with a higher number of axes and spindles than the entry-level NCU variants.
- Supports multi-channel systems where each channel can control multiple interpolating axes and associated spindles (e.g. large machining centers, multi-spindle transfer lines).
- Axis expansion is achieved via NX10.3/NX15.3 or similar expansion modules, which are synchronized with the master NCU, enabling:
 - High-speed 5-axis contouring,
 - Simultaneous machining on multiple heads/turrets,
 - Coordinated multi-spindle and gantry configurations.

High-Speed Processing

Uses a more powerful multi-core CPU compared to lower NCU types, resulting in:

- Very short internal interpolator cycle times,
- High look-ahead depth for complex 3D toolpaths,
- Support for tighter dynamic tolerances and higher path precision.

Optimized for high-speed machining (HSC):

- Advanced jerk-limited motion profiles,
- Fine interpolation of small blocks and splines,
- Improved surface finish and reduced machining time, especially in mold and die applications.

Integrated Safety Technology

- Includes Safety Integrated functions in conjunction with SINAMICS drives, such as:
 - Safe Torque Off (STO),
 - Safe Stop 1 (SS1),
 - Safe Stop 2 (SS2) in appropriate configurations,
 - Safe Limited Speed (SLS), Safe Limited Position (SLP), Safe Direction (SDI), and similar axis-related safety functions (depending on drive configuration and licensing).

- Meets functional safety standards such as EN ISO 13849-1 and IEC 61508 (up to typical PL/SIL levels specified in Siemens documentation).
- Safety logic is engineered in TIA Portal (F-configuration), with certified safety function blocks and diagnostics to validate:
 - Emergency stop reaction,
 - Guard door monitoring,
 - Enabling circuits for set-up/jog modes,
 - Safe operating stop for manual intervention.

Universal Engineering Environment

- CNC, PLC, drives, and safety are configured using:
 - TIA Portal for PLC, safety, PROFINET, and basic hardware topology,
 - SINUMERIK ONE engineering tools (e.g. Create MyVirtual Machine, Startdrive integration) for CNC, kinematics, drives, and technology functions.
- Virtual commissioning is supported via digital twin concepts: the NCU 1760 configuration can be coupled to a virtual machine model to test PLC sequences, CNC programs, and HMI without actual hardware.

2.1.4 Communication Interfaces and Protocols

The NCU 1760 offers a set of Ethernet-based interfaces and real-time fieldbuses similar in principle to the NCU 1740, but sized for larger systems.

PROFINET

Main backbone for drive and I/O communication as well as HMI and cell-level networking:

- Dedicated real-time PROFINET ports for:
 - Motion-critical communication with SINAMICS drives and distributed I/O (RT/IRT).
 - Additional PROFINET IO devices such as I/O racks, safety I/O, and technological field devices (e.g. measuring systems, hydraulic units).
- Typical logical separation (naming for illustration, check exact designation on the NCU 1760 front panel):
 - Motion/Drive network: real-time / isochronous communication with drives,
 - Machine-level I/O network: sensors/actuators, safety I/O,
 - Cell / plant network: communication to line controllers, SCADA, robotic cells, and MES.

Industrial Ethernet

TCP/IP-based interfaces for:

- HMI panels (PPU 17xx, IPCs with SINUMERIK Operate),
- File services (NC program transfer, tool data, offset data, log files),

- Remote access for diagnostics, commissioning, and remote service (via VPN or secure tunnels),
- Interfaces to higher-level systems (ERP/MES, condition monitoring servers, data analytics platforms).

Supports secure communication concepts (e.g. firewalls, VLANs, access control) when integrated into a protected plant IT architecture.

DRIVE-CLiQ

High-speed, deterministic interface for connecting SINAMICS drive components, motors, and encoders:

- Connects to S120/S210 or other supported SINAMICS variants depending on the machine design.
- Provides automatic component recognition, transfer of motor nameplate data, and plausibility checks.
- Ensures synchronized data exchange across all axes in the system, crucial for multi-axis interpolation and gantry synchronism.

Other Interfaces (depending on version)

Optional/provided interfaces may include:

- PROFIBUS DP for compatibility with existing systems or legacy I/O,
- USB/compact flash for service and data backup (where present),
- Service-specific Ethernet port for commissioning, trace, and diagnostics separated from production networks.

2.1.5 Connected Subsystems and Peripherals

Drive System (SINAMICS)

The NCU 1760 is typically paired with SINAMICS S120 or comparable drive systems for high-dynamic multi-axis applications.

Each axis (linear or rotary) is controlled in closed loop via:

- Position, speed, and current control loops in the drive,
- High-resolution encoder feedback via DRIVE-CLiQ,
- CNC interpolation in the NCU with tight synchronization to the drive control cycle.

Supports:

- Standard servo axes, main spindles, auxiliary axes,
- Gantry axes with electronic master-slave relationships,
- Synchronous operation between axes (e.g. for twin-spindle machines).

Sensors, Actuators, and I/O Units

Distributed ET 200 I/O stations and safety I/O modules are connected over PROFINET. Typical field devices:

- Limit switches, reference switches, pressure/temperature sensors,
- Tool changer actuators, clamping systems, coolant and lubrication systems,
- Safety devices such as guard doors, light curtains, and enabling switches.

I/O mapping is defined in both TIA Portal and SINUMERIK configuration to ensure deterministic interaction between CNC and PLC logic.

Human-Machine Interface (HMI)

HMIs such as SINUMERIK PPU 1740 or IPC-based panels with SINUMERIK Operate connect via Ethernet/PROFINET.

Provide:

- Program editing and management (ShopMill/ShopTurn and ISO),
- Online NC, PLC, and alarms diagnostics,
- Jogging, set-up, tool and offset management,
- Visualization of machine status and technology-specific screens.

Additional hardware operator components (MCP, MPP, handwheels) connect via PROFINET or dedicated connections defined in the configuration.

Axis Extension Modules

NX10.3, NX15.3, or similar axis expansion units are used to:

- Increase the total number of axes beyond the onboard capability,
- Distribute drive connections physically closer to machine axes,
- Offload some real-time data handling while remaining fully synchronized with the master NCU 1760.

2.1.6 CNC Configuration Parameters

The configuration principles for the NCU 1760 are analogous to the NCU 1740, but applied to larger/more complex machines.

Axis Configuration

Define all linear and rotary axes, including:

- Axis names, machine directions, stroke limits, soft/hard limit switches,
- Velocity, acceleration, and jerk limits, according to mechanical constraints,
- Interpolation groups for each channel and assignments of axes to those groups.

Configure machine kinematics for complex structures:

- 5-axis table/head combinations,
- Portal and gantry axes,
- Special kinematics (e.g. grinding, gear cutting) where required.

Drive and Motor Setup

Assign each axis to its corresponding DRIVE-CLiQ component (drive object and motor).
Configure drive-related parameters:

- Motor data (rated torque, current, speed, motor constants),
- Current, speed, and position controller tunings,
- Limit values for torque, speed, and power.

Perform automatic and manual optimization:

- Automatic tuning wizards in SINAMICS/SINUMERIK engineering,
- Verification of stability, following error, and dynamic response via trace functions.

Encoder and Feedback Systems

Define encoder types and resolutions for each axis:

- Motor encoder (primary),
- Measuring system on the machine bed or ball screw (secondary, if used).

Configure:

- Absolute vs. incremental encoders,
- Reference procedures and homing modes,
- Compensation for backlash, pitch errors, and mechanical compliance using CNC compensation tables.

PLC Control Configuration

The internal S7-1500F PLC handles:

- Basic machine functions: hydraulics, pneumatics, lubrication, spindle orientation, auxiliary axes,
- Tool changer sequences, pallet changers, workholding, and safety-related interlocks.

Configuration steps:

- I/O mapping in hardware configuration (PROFINET topologies, address assignment),
- PLC program structure (organization blocks, cyclic tasks, safety program sections),
- Interface signals between PLC and CNC (HMI signals, mode control, M-functions, channel enabling).

Safety Configuration

Safety functions are realized by the F-PLC and drive-based Safety Integrated:

- E-stop channels, guard door monitoring, enabling buttons, safety light curtains, etc.
- Safe speed and position monitoring for set-up modes, tool change, and maintenance.

Engineering tasks:

- Configure F-I/O modules and assign safe addresses,
- Implement safety logic with certified function blocks in the F-program,
- Parameterize drive-based safety functions (e.g. SLS limits) in SINAMICS.

Validation and documentation:

- Perform safety acceptance tests according to the required PL/SIL,
- Document safety functions, reaction times, and test procedures as required by standards and your machine-builder process.

2.2 MCP Configuration and Technical Integration

[3]

The technical integration of a Machine Control Panel (MCP) within the Create MyVirtual Machine environment is based on aligning the virtual hardware parameters with the TIA Portal PLC project. For an MCP 483 or a custom User MCP, the connection is established through absolute addressing of inputs and outputs.

2.2.1 Address Mapping and PLC Synchronization

The primary communication mechanism between the virtual SINUMERIK unit and the PLC is the configuration of start addresses within the MCP Editor. These addresses must correspond exactly to those defined in the TIA Portal project to ensure that the PLC user program processes signals correctly.

- **Standard I/O Addressing:** Absolute byte addresses must be assigned for both MCP inputs and MCP outputs.
- **Default Parameters:** The default starting addresses are %I0.0 for inputs and %Q0.0 for outputs.
- **Address Range:** Permissible byte address values range from 0 to 8192.
- **Safety Integration:** The EMERGENCY STOP requires a dedicated bit-level address (e.g., 9.0) for both input and output channels to ensure proper integration with the safety logic.

2.2.2 Configuration Requirements

For the data link to be established or modified, the following system conditions must be satisfied:

- **Operational State:** The MCP configuration is editable only when the machine project is open and the simulation is stopped.

- **Memory Persistence:** Configuration changes, such as modifications to start addresses, are finalized and written to the machine project only after a manual save operation.
- **Non-Editable Hardware:** Certain hard-wired components, including the EMERGENCY STOP, feedrate and spindle overrides, and keyswitches, have fixed functional mappings that cannot be modified within the editor.

2.2.3 Address Verification

The integration interface provides immediate validation of technical parameters:

- **Range Validation:** If a byte address exceeds the permissible range of 0–8192, the corresponding field is highlighted in red and the value is rejected.
- **Bit-Level Formatting:** The system enforces the <byte address>.<bit number> format for all EMERGENCY STOP configurations.

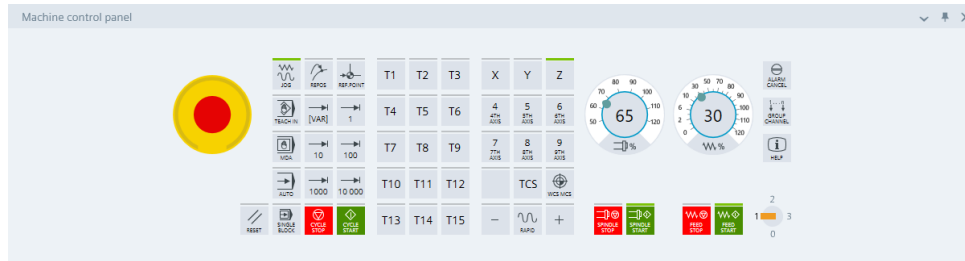


Figure 2.1: MCP in Virtual Machine

2.3 Technical Overview of SINAMICS S120 Motor Modules in CNC Architecture

This section analyzes the technical specifications and functional roles of the Siemens SINAMICS S120 Motor Modules—specifically models **6SL3120-1TE13-0AD0** and **6SL3120-2TE13-0AD0**—within a CNC control panel. As core components of the Booksize D-Type series, these modules operate as power conversion units within a modular, multi-axis drive framework, facilitating high-performance motion control through shared DC-link architecture.

2.3.1 Functional Principles of the S120 Drive System

In the SINAMICS S120 ecosystem, Motor Modules function as the power stage, converting DC-link voltage into a pulse-width modulated (PWM) variable three-phase AC voltage to actuate motor movement. While the central Control Unit (CU) manages the sophisticated motion logic and command execution, the Motor Modules execute the physical power conversion. Integration is achieved via a common DC busbar for energy efficiency and the **DRIVE-CLiQ** digital interface for high-speed communication.

2.3.2 Comparative Analysis of Module Specifications

The following table delineates the primary technical differences between the Single and Double Motor Module variants used in this application:

Table 2.1: Technical Comparison of SINAMICS S120 Motor Modules

Parameter	6SL3120-1TE13-0AD0	6SL3120-2TE13-0AD0
Module Configuration	Single Motor Module	Double Motor Module
Inverter Count	1	2
Rated Current (I_n)	3 A	2×3 A
Axis Control Capacity	1 Axis	2 Axes
Cooling Method	Internal Air Cooling	Internal Air Cooling
DC Link Voltage Range	510 V – 720 V DC	510 V – 720 V DC
Communication Protocol	DRIVE-CLiQ	DRIVE-CLiQ
Dynamic Performance	D-Type (High-Dynamic)	D-Type (High-Dynamic)
Protection Rating	IP20	IP20

2.3.3 Strategic Application within CNC Control Circuits

The selection of these modules is dictated by the torque and power requirements of the CNC axes. The **6SL3120-1TE13-0AD0** (Single Module) is typically reserved for the primary spindle or heavy-duty feed axes that demand dedicated power throughput. Conversely, the **6SL3120-2TE13-0AD0** (Double Module) provides a space-efficient solution for auxiliary servo axes, such as rotary tables or tool changers, by housing two independent inverters within a single chassis. This modularity reduces the physical footprint of the control cabinet and simplifies inter-module wiring.

2.3.4 Motor Compatibility and Control Interfacing

The S120 series supports a diverse range of Siemens motors, with control modes (position, speed, or torque) defined during the parameterization of the Control Unit. Compatible motor series include:

- **Synchronous Servo Motors:** 1FK7 and 1FT7 series, optimized for high-precision CNC feed axes.
- **Asynchronous Induction Motors:** 1PH8, 1MB, or 1LA series, typically utilized for spindle drive applications.

2.3.5 Network Integration and Diagnostic Communication

System-wide synchronization is maintained through the **DRIVE-CLiQ** interface, a proprietary backplane that facilitates the transfer of feedback, diagnostics, and electronic rating plate data. For higher-level integration, the Control Unit interfaces with the CNC or PLC via **PROFINET** or **PROFIBUS**, ensuring deterministic real-time motion control and centralized diagnostic capabilities across all axes.

2.3.6 Representative CNC System Configuration

A standard implementation within a CNC machine tool environment typically utilizes the following topology:

- **Primary Drive:** 6SL3120-1TE13-0AD0 allocated to the main spindle or a high-torque axis.
- **Secondary Drives:** 6SL3120-2TE13-0AD0 managing two smaller auxiliary servo axes.

This configuration is powered by a common **Line Module** (e.g., 6SL3130 series), which enables regenerative braking and energy exchange between motoring and generating axes, significantly enhancing the overall energy efficiency of the CNC system.

3 Methodology and Workflow

3.1 System Overview and Engineering Architecture

The engineering methodology employed in this case study is grounded in the digital twin concept, wherein a SINUMERIK ONE CNC controller is fully commissioned and validated within a software-based virtual environment prior to any interaction with physical hardware. This approach, facilitated by the Create MyVirtual Machine (CMVM) simulation platform developed by Siemens, enables parallel development of the Programmable Logic Controller (PLC) program, Numerical Control (NC) machine data, Human–Machine Interface (HMI) all within a risk-free and cost-effective simulation context.

The engineering process is structurally bifurcated into two distinct phases: the virtual machine phase, in which the SINUMERIK ONE controller is commissioned entirely within CMVM, and the real machine phase, wherein the validated commissioning data are transferred and adapted to the physical SINUMERIK ONE hardware. In this case study, the scope is confined to the virtual machine phase, covering the complete workflow from project creation in TIA Portal through NC axis configuration, HMI verification, and Integrated programming.

3.1.1 Architecture of SINUMERIK ONE CNC system

Introduction

The SINUMERIK ONE CNC System represents Siemens’ latest generation of integrated machine control technology. It merges CNC control, automation logic, drive regulation, and operator interface functions into one cohesive digital platform. This integration simplifies the overall machine architecture, increases reliability, and reduces commissioning time by allowing all disciplines—motion control, logic control, safety, and visualization—to operate under a unified environment.

In essence, SINUMERIK ONE is designed as the “digital native” CNC, seamlessly bridging the physical and virtual worlds of machine tool engineering through its compatibility with the Digital Twin environment. Figure 1 illustrates the simplified data flow and control interaction within this architecture.

Main Control Units

At the core of this architecture are two principal control components:

SIMATIC S7-1500 PLC Acts as the machine controller, managing the general automation and safety functions across the entire system. It executes machine logic, input/output management, safety interlocks, and auxiliary control tasks such as coolant operation, lubrication, and tool changer coordination. Being a fail-safe CPU variant, it can integrate both standard control logic and safety-related functions on the same hardware platform.

NCU 1760 Numerical Control Unit Operates as the motion and path control unit of the system. It computes all axis interpolations, spindle dynamics, and kinematic transformations in real time. Through high-performance motion processors, it ensures synchronized multi-axis movement and precise execution of CNC part programs. The NCU and the PLC coexist within the same SINUMERIK ONE environment, sharing data and synchronization signals via fast backplane communication.

Together, these components ensure tight coordination between machine automation logic and precision machining control, enabling smooth interaction between all subsystems.

Drives and Feedback System

The NCU 1760 communicates directly with SINAMICS drive modules, which supply power to the respective machine axes and spindles. Each axis drive executes motion commands generated by the NCU while continuously feeding back real-time position and speed data through DRIVE-CLiQ communication.

This closed-loop feedback process ensures that actual motion precisely follows the programmed tool path, thereby maintaining machining accuracy and repeatability. The system supports an array of motor types—servo, spindle, and linear—and relies on feedback encoders to detect positional deviations. Adjustments are calculated dynamically, ensuring stable performance even during fast axis movements or high-load conditions.

The high-speed data exchange between the NCU and SINAMICS S120 drive components guarantees stable and deterministic motion responses critical for precision machining operations.

Operator Interface

The Human-Machine Interface (HMI) connects the operator to the control system through a direct link to the NCU/S7-1500F unit. It facilitates all essential interactions including mode selection, manual jogging, program start/stop, and visualization of machine states. Siemens provides a range of compatible HMI devices such as the Machine Control Panel (MCP) and Machine Pushbutton Panel (MPP).

Operator input data (e.g., feed override, mode switch, or program execution) interact in real time with drive feedback data from encoders and position sensors. This synchronized relationship ensures that human commands translate into effective machine action with accurate response and safety monitoring.

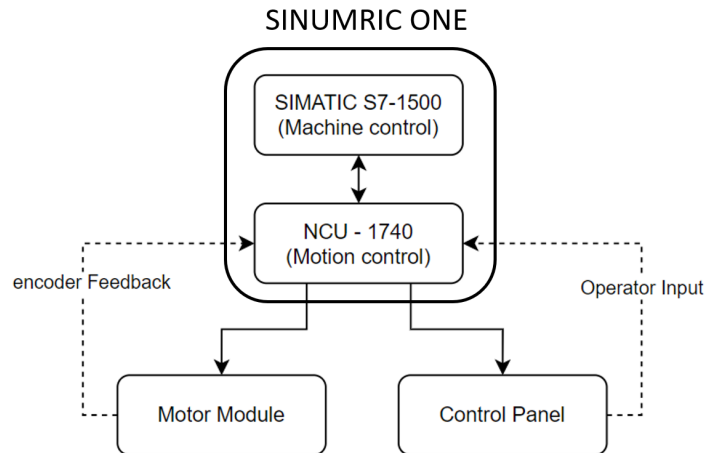


Figure 3.1: Interconnections between PLC, NCU, Drive Modules, and HMI within a SINUMERIK ONE system.

3.1.2 Component-wise Engineering Flow (PLC → NC → HMI)

The engineering workflow follows a component-sequential structure, progressing through four principal subsystems of the SINUMERIK ONE controller. Each component is configured, compiled, and downloaded into CMVM independently before the subsequent layer is addressed, ensuring modular verification at each stage.

The sequence initiates with PLC engineering in TIA Portal V20, encompassing hardware configuration of the NCU, IP address assignment, activation of block simulation capability, integration of the basic PLC program (LBP), and development of the user PLC program. Upon

successful loading of the PLC configuration into the virtual machine, NC commissioning is performed through SINUMERIK Operate—the embedded HMI—by configuring machine data for axis naming, axis types, and spindle assignment. The HMI layer requires no independent commissioning when the standard SINUMERIK Operate interface is used, as it is automatically instantiated by CMVM upon CNC software selection.

3.1.3 Software Stack Used

TIA Portal V20, SINUMERIK ONE STEP 7 Toolbox V20, Create MyVirtual Machine V1.5.1.1, CNC SW V6.25 [web:1][web:4].

The simulation environment was established using a single-PC installation, wherein all required engineering and simulation tools were co-hosted on the same system, communicating via Ethernet using PROFINET/Industrial Ethernet protocols. The complete software stack employed in this case study is detailed in Table 1.1.

Table 3.1: Software components and versions used in the virtual CNC machine simulation

Software Component	Version Used	Primary Role
SIMATIC STEP 7 Professional (TIA Portal)	V20	PLC hardware configuration and programming environment
SINUMERIK ONE STEP 7 Toolbox	V20	Option package for SINUMERIK NCU hardware and basic PLC program
Create MyVirtual Machine	V1.5.1.1	Virtual SINUMERIK ONE controller framework and simulation host
SINUMERIK Virtual CNC Software	V6.25	Virtual simulation of NCK, PLC, HMI, and drive replacement components

Version compatibility between the CNC software and the TIA Portal project is a critical prerequisite. As specified in the commissioning documentation, CNC SW V6.25 must be paired with the SINUMERIK ONE STEP 7 Toolbox V20, with the basic PLC program also at version V6.25. The SINUMERIK Virtual CNC software package encompasses the simulation kernels for the NCK, PLC, HMI, and a drive replacement component, collectively enabling a functionally complete virtual controller environment.

3.1.4 3.1.4 Machine Configuration Used (3 linear axes: X1, Y1, Z1)

The final virtual machine configuration used in this case study includes three linear axes only (X, Y, Z), mapped as machine axes MX1, MY1, and MZ1 with channel axis names X1, Y1, and Z1. A spindle and additional rotary axes were initially considered; however, spindle commissioning in the CMVM environment required an additional license in the laboratory setup, therefore the spindle was not included in the final commissioned configuration and only the three linear axes were validated.

The machine control panel (MCP) used in the virtual environment is the MCP483 Milling version, configured as an Ethernet MCP within the PLC program at function block FC1. The MCP interface enables direct operator interaction with the simulated controller during functional testing, providing axis traversal commands, mode selection, and feedrate override control. This configuration forms the operational baseline for all subsequent commissioning steps described in the following sections.

Table 3.2: Axis configuration for the virtual CNC machine simulation (3 linear axes)

Machine Axis	Axis Identifier	Channel	Axis Name	Axis Type
Machine Axis 1	MX1	X1		Linear
Machine Axis 2	MY1	Y1		Linear
Machine Axis 3	MZ1	Z1		Linear

3.2 Setting Up the Virtual Environment

3.2.1 Installation Environment

Single-PC setup (TIA Portal + Create MyVirtual Machine on same system).

In this case study, all engineering and simulation software was deployed on a single workstation, eliminating the need for cross-computer Ethernet firewall configuration. The mandatory installation sequence was strictly observed: TIA Portal V20 with the SINUMERIK ONE STEP 7 Toolbox V20 was installed first, followed by Create MyVirtual Machine (CMVM) V1.5.1.1, and finally the SINUMERIK Virtual CNC Software V6.25. This sequence is critical because CMVM must detect the TIA Portal installation during its own setup process. Both tools communicate via the local Ethernet stack; when co-hosted on the same machine, the local host IP address (e.g., 127.0.0.1) is used during the download phase rather than a physical network adapter address.

3.2.2 Creating a Machine Project in Create My Virtual Machine

CMVM organises all controller data — NC, HMI, PLC, and drive parameters — within a structured file format called a machine project, stored with the *.vcp (Virtual Commissioning Project) extension. Before any TIA Portal configuration can be transferred to the virtual controller, a machine project must first be created and opened within CMVM. The machine project was created by launching CMVM and selecting the option to create a new project. CNC SW version V6.25 was selected from the drop-down list of installed CNC software versions, which determines the SINUMERIK component versions instantiated in the simulation. Project settings were then configured, specifying the HMI display resolution appropriate for the available monitor and selecting the MCP 483 Milling as the control panel type — the same panel used for the virtual machine in this configuration. Upon confirmation, the project view opened in an initially empty state, containing a standard default controller configuration with no user-defined PLC program or NC data loaded.

3.2.3 Starting the Machine Simulation

With the machine project open, the virtual machine was initiated by clicking the "Start machine" button in the CMVM simulation control area. This button remains inactive (grayed out) until CMVM has fully loaded the project; it becomes available once the environment is ready. The startup process replicates the power-on and ramp-up sequence of a physical SINUMERIK ONE controller. Successful initialisation is confirmed when all virtual components — NCK, PLC, HMI, and drive replacement — display a green status indicator in the component overview panel.

Since no PLC program had yet been downloaded at this stage, the virtual controller reported a PLC error upon startup, which is the expected behaviour for an empty machine project. This state serves as the baseline from which PLC and NC commissioning proceed in subsequent steps.

Saving the machine project requires the simulation to be in Stop status. Changes are not saved automatically; the project must be explicitly saved via "Save" or "Save as" in the menu bar, after which it is written to the *.vcp file. On subsequent openings, CMVM resumes from the last saved state.

3.2.4 Configuring Network Communication between TIA Portal and Create MyVirtual Machine

TIA Portal V20 and CMVM V1.5.1.1 communicate via an Ethernet connection using the PROFINET/Industrial Ethernet protocol. In a single-PC setup, this communication is handled entirely through the local network stack without requiring a dedicated physical network interface between two separate systems. However, the network adapter used for the CMVM connection must have its DCP (Discovery and Configuration Protocol) checkbox appropriately configured under CMVM's Communication Settings (LLDP/DCP settings). If an incompatible device is detected during TIA Portal's device search, the DCP checkbox for the relevant adapter must be unchecked within CMVM to resolve the communication conflict.

3.2.5 IP Address Configuration and PROFINET Setup

The IP address for the Communication Processor (CP) at interface X150 of the SINUMERIK NCU was configured within the TIA Portal project under the hardware device properties. This address establishes the PROFINET identity of the virtual NCU and is the target address used by TIA Portal when searching for devices during the hardware download procedure. In the single-PC configuration used here, the IP address of the local host (192.168.0.1) may alternatively be entered directly during the download dialog, bypassing the need to pre-assign an IP address in the hardware configuration of the CP. The automatically generated PROFINET device name assigned by TIA Portal was retained without modification. This network configuration forms a prerequisite for all subsequent hardware and software downloads from TIA Portal into the running CMVM virtual machine.

3.3 Engineering the PLC

3.3.1 Creating the TIA Portal Project

The PLC engineering phase was initiated entirely within TIA Portal V20, which serves as the sole engineering environment for hardware configuration and PLC program development of the SINUMERIK ONE controller. The availability of SINUMERIK NCU device families within TIA Portal is contingent on the prior installation of the SINUMERIK ONE STEP 7 Toolbox V20, which registers the necessary hardware catalogue entries and function libraries into the TIA Portal environment.

Project Creation

A new project was created by selecting "Create new project" from the TIA Portal portal view. A project name was assigned and a storage path was defined before confirming with "Create". Upon creation, the environment was switched from the portal view to the full project view, which provides access to the project navigator, hardware and network editors, and the PLC programming workspace. At this stage, the project contained no devices and no program blocks — it constituted a blank workspace in which all subsequent hardware and software configuration was to be built.

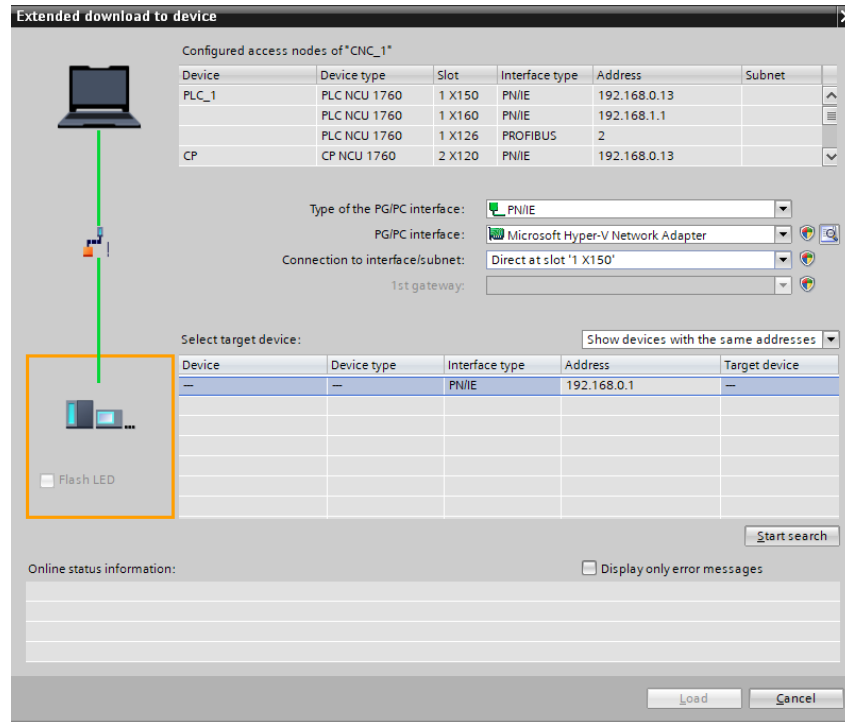


Figure 3.2: Extended Load dialog showing PN/IE interface settings and target device selection, alongside the Load Preview dialog with configuration summary

NCU Hardware Insertion

The SINUMERIK NCU was inserted into the project by selecting "Add new device" from the project navigation panel and navigating to the "Controllers" category. Within the folder structure, the SINUMERIK ONE device family was expanded to reveal the available NCU variants. The appropriate NCU model was selected, and the firmware version was set to V6.25 in the version drop-down list, ensuring consistency with the CNC SW V6.25 installed in CMVM V1.5.1.1. The selection was confirmed with "OK", after which the NCU appeared as a new device in the project tree.

Structure of the Inserted NCU

Upon insertion, the SINUMERIK NCU was automatically populated with all its integrated subcomponents as an indivisible unit. These subcomponents — the PLC (SIMATIC S7-1500), NCK (Numerical Control Kernel), CP (Communication Processor), HMI (SINUMERIK Operate), and SINAMICS Integrated (drive interface) — are permanently bound to the NCU object and cannot be individually copied, moved, or deleted within the project. This architectural constraint ensures that the NCU is always handled as a coherent system entity. For the machine configured in this case study comprising three linear axes (MX1, MY1, MZ1), no additional NX extension modules were required, as the standard NCU accommodates up to six axes without supplementary hardware.

3.3.2 Configuring the PLC Hardware

With the NCU successfully inserted into the TIA Portal V20 project, the hardware configuration required several specific settings to be applied before the project could be compiled and downloaded into the CMVM virtual environment.

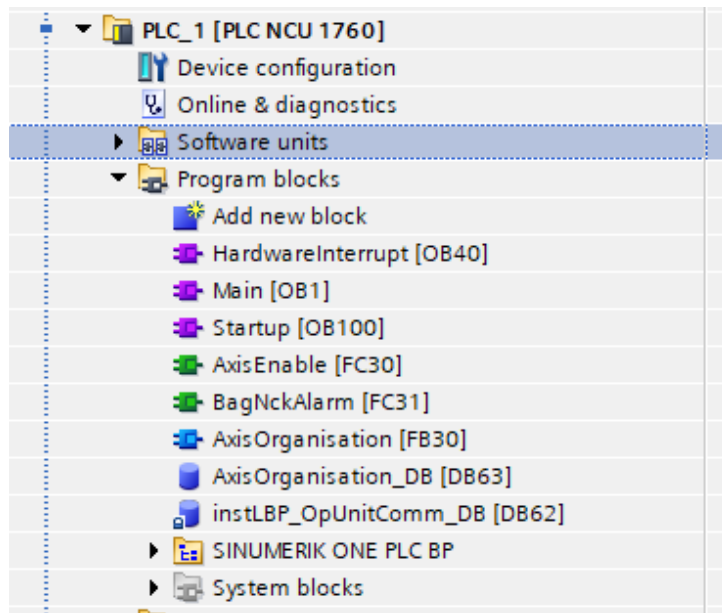


Figure 3.3: TIA Portal V20 project view showing the empty project tree after creation

Setting the IP Address (X150 of CP)

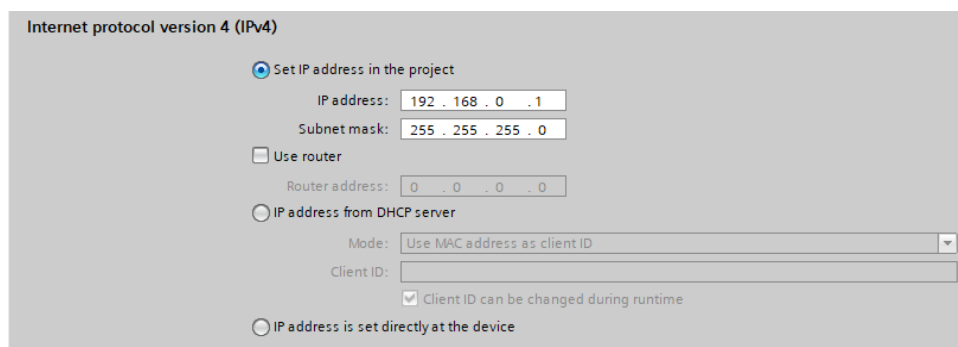


Figure 3.4: TIA Portal hardware device view showing Ethernet interface X150 properties with IP address configuration

Activating Simulation-Capability of Blocks

A prerequisite specific to the virtual machine workflow — absent in real machine commissioning — is the activation of block simulation capability within the TIA Portal project properties. This setting is disabled by default, as enabling it can affect know-how protection mechanisms applied to function blocks. Without this activation, blocks loaded into CMVM fail the simulation check during download.

To enable this setting, the project root was right-clicked in the project tree and "Properties" was selected. Under the Protection tab, the option "Support the simulation-capability of blocks for compilation" was activated and confirmed with "OK". Once enabled, each block is verified for simulation compatibility at compile time. Should any block fail this check — for instance, a know-how protected block sourced from an incompatible library — a specific error message is displayed identifying the offending block, and the block must either be replaced with a simulation-compatible version or excluded from the project.

Activating Message Reception for PLC

To permit diagnostic messages generated within the virtual PLC to be displayed in TIA Portal after downloading, the message reception setting was enabled for the PLC device. This was performed by double-clicking the "Online & Diagnostics" folder of the PLC in the project navigation, selecting the "Online accesses" group in the area navigation panel, and activating the "Receive messages" option. This setting only takes effect after the online connection to the virtual device is re-established following the download; messages generated prior to the next online connection are not retroactively captured.

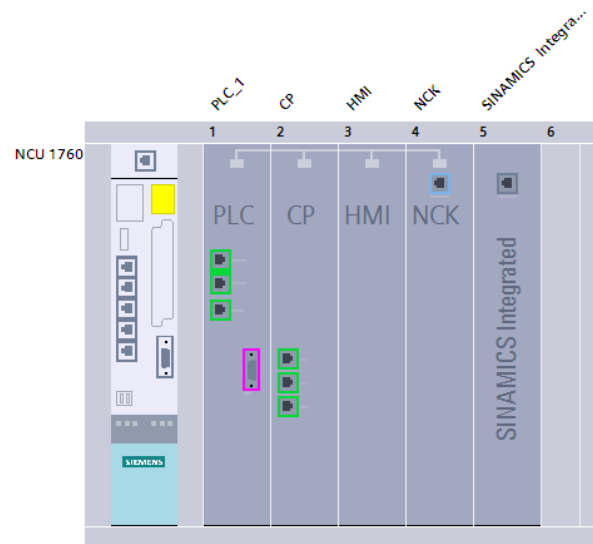


Figure 3.5: TIA Portal hardware configuration view showing the fully configured NCU with CP interface X150 settings and project protection properties

3.3.3 Compiling and Loading to the Virtual Machine

Compiling the Hardware Configuration

Before the hardware configuration could be transferred to CMVM V1.5.1.1, it was necessary to compile the project within TIA Portal V20. Compilation was initiated by right-clicking the "CNC_1" entry in the project tree and selecting "Hardware (changes only)" from the Compile shortcut menu. During this process, TIA Portal performs a consistency check on the entire hardware configuration, validating the NCU, all its integrated subcomponents — PLC, NCK, CP, HMI, and SINAMICS Integrated — as well as any additionally connected optional components. Any configuration errors detected at this stage must be resolved before compilation can complete successfully. A clean compilation without errors is a mandatory prerequisite for the subsequent download step; the process cannot proceed to device loading until the hardware configuration is in a fully consistent state.

Loading Hardware Configuration into Create MyVirtual Machine via PN/IE

Prior to initiating the download, the following conditions were verified:

- the simulation-capability of blocks had been activated
- the IP address at interface X150 of the CP had been configured in the project

- the hardware configuration had been compiled without errors
- CMVM V1.5.1.1 was running with the machine project open and the simulation started
- the CNC SW version (V6.25) was confirmed to be compatible with the SINUMERIK ONE STEP 7 Toolbox V20

The hardware configuration was loaded into the virtual machine by right-clicking the NCU in the project navigator and selecting *"Load to device > Hardware configuration"*. This opened the Extended Load dialog, in which the following interface parameters were configured:

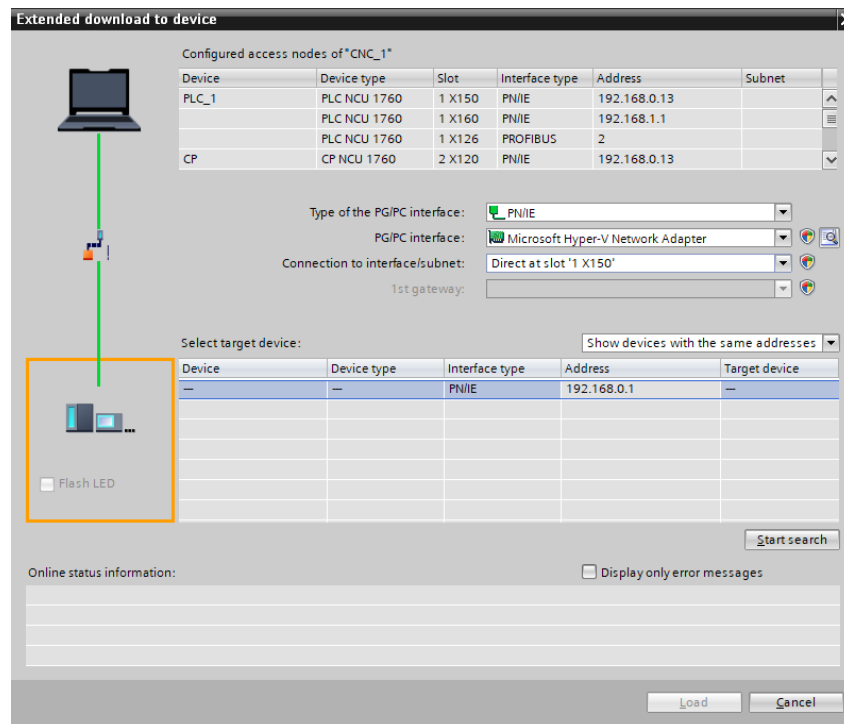


Figure 3.6: Extended Load dialog showing PN/IE interface settings and target device selection, alongside the Load Preview dialog with configuration summary

- **PG/PC interface type:** PN/IE
- **PG/PC interface:** The active network adapter on the host machine
- **Connection with interface/subnet:** Directly at slot X150
- **Select target device:** Display devices with the same addresses

"Start search" was then clicked to initiate device discovery. TIA Portal scanned the network and returned the virtual SINUMERIK NCU running within CMVM as a detectable target device. The appropriate device was identified by its IP address in the list of compatible nodes.

As an alternative to pre-configuring the IP address in the hardware properties, the address can be entered directly in the *"Address"* column of the compatible nodes list during this step, after which the search starts immediately without needing to click *"Start search"* separately.

Load Preview and Confirming the Download

After selecting the target device and confirming with *"Load"*, TIA Portal presented the Load Preview dialog. This dialog summarises all configuration objects scheduled for transfer

to the target device, allowing verification of the intended changes before any data is written to the virtual controller.

The preview was reviewed to confirm that the hardware configuration objects corresponded to the intended NCU setup for three linear axes (MX1, MY1, MZ1) only.

Following confirmation with "Load" in the preview dialog, the transfer was executed and the "Results of the load procedure" dialog appeared. At this point, an option was presented to restart the PLC upon completion.

Critical note: The PLC cannot be started at this stage in isolation, as no basic PLC program had yet been loaded — the PLC program download constitutes a separate subsequent step. The dialog was confirmed with "*Complete*" to finalise the hardware configuration transfer.

3.3.4 Integrating the Basic PLC Program

Importing the Basic PLC Program (LBP blocks)

The SINUMERIK ONE PLC basic program (LBP) constitutes the foundational software layer that orchestrates signal and data exchange between the PLC user program and the NCK, HMI, and machine control panel components. It is not created manually but is instead sourced from the SINUMERIK ONE PLC Basic Program system library bundled with the SINUMERIK ONE STEP 7 Toolbox V20, accessed through the Libraries task card in TIA Portal V20.

Before the program blocks themselves could be imported, the associated PLC data types (UDTs) were first copied into the project. These data types define the structure of the interface data blocks — covering axes, spindles, channels, mode groups, alarms, tool management, and MCP signals.

The UDT folder "Master copies > SINUMERIK ONE PLC Basic Program > SINUMERIK ONE PLC BP data types" was dragged and dropped into the "PLC data types" folder under the NCU device in the project tree. Unused data types were subsequently removed to avoid extending compilation and archiving times unnecessarily.

With the data types in place, the PLC basic program blocks were imported by dragging the "SINUMERIK ONE PLC BP" copy templates folder from the library directly into the Program blocks folder of the project. This operation populated the project with all block types required — function blocks (FB), functions (FC), data blocks (DB), and internal blocks — organised into corresponding subfolders.

Following the import, the program blocks were compiled via "Compile > Software (compile all blocks)" from the context menu of the Program blocks folder, which also verified the absence of duplicate block addresses across the project.

For the three-axis configuration used in this simulation (MX1, MY1, MZ1), surplus axis data blocks (e.g., LBP Axis4 and higher, if generated during import) and unused channel data blocks were deleted to keep the project lean and consistent.

Overview of Key Function Blocks

LBP_Start [FC1], AxisEnable [FC30], AxisOrganisation [FB30], BagNckAlarm [FC31], Main [OB1].

The PLC program structure follows a hierarchical calling sequence governed by three organisation blocks (OBs), within which the basic program blocks are invoked at defined execution points. The key blocks relevant to this simulation are described below.

LBP@ConfigBP [FC1]

Called exclusively within **Startup [OB100]**, which executes once when the PLC transitions from STOP to RUN. This function configures and initialises the entire basic PLC program, including address mapping for the MCP483 Milling control panel with input address %I0.0

and output address %Q0.0, channel configuration, and axis interface initialisation. It internally invokes further basic program blocks required for complete initialisation.

Main [OB1]

The cyclic organisation block executed continuously during PLC runtime. It contains the call to **LBP_MainBP [FC2]**, which processes the complete NCK–PLC interface cyclically and activates sign-of-life monitoring between PLC and NCK after the first OB cycle.

Additionally, LBP MCPCtrlMilling FC19 is called within Main OB1 to manage signal transmission between the NC/PLC interface and the MCP483 Milling panel, configured with BAGNo B161 and ChanNo B161; spindle-related parameters were not used because no spindle was commissioned in the final virtual configuration. The user-defined blocks **AxisOrganisation [FB30]** and **BagNckAlarm [FC31]** are also called within Main [OB1].

AxisEnable [FC30]

A user-created function that sets the controller enable, pulse enable, feedrate override activation, and position measuring system enable for each individual axis. It is parameterised with the respective axis interface data block (e.g., **LBP_Axis1** for MX1) and is called iteratively within **AxisOrganisation** for each configured axis.

AxisOrganisation [FB30]

A user-created function block that aggregates the axis enable calls for all active axes. For the three linear axes configured in this simulation, three networks were programmed—one per axis—each calling **AxisEnable FC30** with the corresponding axis interface data block. A fifth network within this block activates the feedrate override for channel 1 via the variable "**LBP_Chan1.A_FD_ORA**". The block is instantiated in **Main [OB1]** with **Enable** set to true.

BagNckAlarm [FC31]

A user-created function that defines the controller response to emergency stop activation and interrupt events. It contains three logical networks:

1. a mode group reset network triggered when a channel reset is active
2. an emergency stop activation network that engages "**LBP_NC.A_EMERGENCY**" when the E-Stop input reads zero
3. an acknowledgement network that clears the emergency stop state when the mode group has been reset and the E-Stop button is no longer pressed

3.3.5 Writing the User PLC Program (Sample Program)

Configuring MCP Signals via LBP_MCPCtrlMilling [FC19]

Prior to configuring MCP signal transmission in the main OB, a dedicated variable table was created in the "PLC variables" folder of the project navigator, named "MCP". Into this table, the basic PLC program data types "**LBP_typeMcp483MillingInput**" and "**LBP_typeMcp483MillingOutput**" were dragged from the PLC data types folder. These were assigned the symbolic names **MCPIn** at address %I0.0 and **MCPOut** at address %Q0.0 respectively.

Two additional Boolean variables were added to this table for the hardware emergency stop circuit: **EStop** at address %I9.0 and **EStopLED** at address %Q19.0. These address assignments are specific to the virtual machine environment and would require adaptation for a real hardware configuration.

Within Main [OB1], the function LBP_MCPCtrlMilling [FC19] was inserted by drag and drop from the program blocks folder. This function manages the bidirectional transfer of signals between the NC/PLC interface and the MCP483 Milling control panel. The following input/output parameters were configured, with all other parameters left at their default values:

Table 3.3: LBP_MCPCtrlMilling [FC19] Parameter Configuration

Input/Output	Value	Description
BAGNo	B#16#1	Mode group 1 selected for signal transmission
ChanNo	B#16#1	Channel 1
FeedHold	"LBP_AlarmMsgs.Chan[1].FDD[0]"	Feed disable signal for channel 1

Programming Axis Enable Logic in LAD

The axis enable logic was implemented across two user-created blocks working in tandem: AxisEnable [FC30] and AxisOrganisation [FB30].

AxisEnable [FC30] contains four LAD networks (one for each enable signal type), each consisting of a normally open (NO) contact connected to the Enable input variable and an assignment coil targeting the corresponding axis signal in the axis interface data block AxisDB. The four enable signals activated per axis were:

- override activation (`AxisDB.A_ORActive`)
- pulse enable (`AxisDB.A_PulseEnable`)
- controller enable (`AxisDB.A_ContrEnable`)
- position measuring system enable (`AxisDB.A_PosMeas1`)

The block was declared with two interface variables: an input **Enable** of type Bool, and an InOut **AxisDB** of type `LBP_typeAxisX`, which carries the full axis/spindle signal interface.

AxisOrganisation [FB30], AxisEnable [FC30] was called once per configured axis. For the three linear axes, three networks were programmed. Each network contains:

1. an NO contact checking whether the axis is actively configured via `[LBP@ConfigData.ActiveAxis]`
2. a second NO contact for the Enable input
3. a call to AxisEnable [FC30] with the corresponding axis data block:
 - `LBP_Axis1` for MX1
 - `LBP_Axis2` for MY1
 - `LBP_Axis3` for MZ1

Programming Emergency Stop Reaction

The emergency stop logic was encapsulated within BagNckAlarm [FC31] and comprised three independent LAD networks. The first network implements mode group reset: two NO contacts are placed in series – one connected to `"LBP_Chan1".A_Reset` (indicating an active reset in channel 1) and one to `[LBP@ConfigData.ActiveChan]` (confirming channel 1 is active). When both conditions are satisfied, the coil `"LBP_ModeGroup".A_MGReset` is set, resetting the mode group. The second network implements emergency stop activation: a normally closed

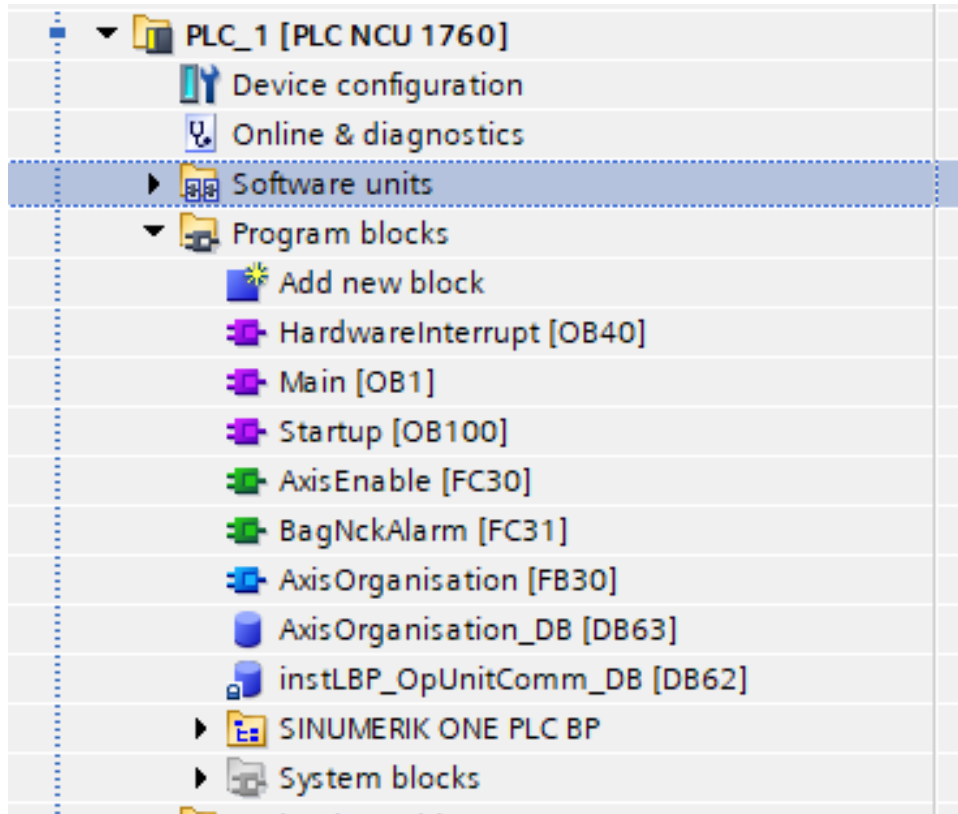


Figure 3.7: TIA Portal LAD editor showing AxisEnable [FC30] networks with enable signal assignments, and AxisOrganisation [FB30] showing the three axis-specific AxisEnable call networks

(NC) contact connected to "EStop" drives the assignment coil "LBP_NC".A_EMERGENCY. When the E-Stop button is pressed, the EStop variable reads 0, which satisfies the NC contact condition (value 0 fulfils the condition), setting the emergency stop flag and braking all axes immediately. The third network implements emergency stop acknowledgement: two NO contacts in series – "LBP_ModeGroup".A_MGReset and "EStop" – must both be true simultaneously, meaning the mode group has been reset and the E-Stop button is no longer pressed. When satisfied, the coil "LBP_NC".A_EMERGENCY_Ackn is activated, clearing the emergency stop state and returning the machine to a ready condition.

Activating Feedrate Override for Channel 1

The feedrate override activation was programmed as an additional network within AxisOrganisation [FB30], appended after the four axis enable networks. This network consists of a single NO contact connected to the Enable input variable, with the assignment coil targeting "LBP.Chan1".A_FD.ORA – the feedrate override activation flag for channel 1. When the Enable input is true, the feedrate override for the channel becomes active, permitting axis traversal commands from the MCP to modify the programmed feed rate. Since only one channel is configured in this simulation, a single such network was sufficient; additional channels would each require a dedicated override activation network.

Following completion of all user program blocks, the entire PLC software was compiled via "Compile > Software (changes only)" from the PLC context menu in the project navigator, after which the complete project – hardware and software – was loaded into the running CMVM virtual machine.

3.4 Engineering the NC (Numerical Control)

With the PLC hardware configuration and user program successfully downloaded into CMVM V1.5.1.1, NC commissioning was initiated through the SINUMERIK Operate interface running within the virtual machine. All NC parameterisation is performed directly via the Operate HMI — not through TIA Portal — by navigating to the Setup operating area and accessing the machine data editors from there. A minimum protection level of 2 (service) or higher must be set in the Setup area before any machine data modifications can be made.

3.4.1 Machine Data Structure

The SINUMERIK ONE NC is parameterised through a structured hierarchy of machine data (MD) and setting data (SD). Machine data govern the fundamental behaviour of the control system and are subdivided into the following categories, each occupying a defined numerical address range:

Table 3.4: Machine data categories by MD number range

MD Range	Category
10000 – 18999	General NC machine data
20000 – 28999	Channel-specific machine data
30000 – 38999	Axis-specific machine data
51000 – 51299	General configuration machine data
52000 – 52299	Channel-specific configuration machine data
53000 – 53299	Axis-specific configuration machine data

Setting data (SD) complement the machine data and are further categorised into general (41000–41999), channel-specific (42000–42999), and axis-specific (43000–43999) ranges. Unlike machine data, changes to setting data take effect immediately without requiring a reset.

Each machine data entry carries an effectiveness attribute that defines when the entered value becomes active in the controller: po (power on / NCK Reset required), re (activated by pressing the RESET key), cf (activated via the "Apply MD" softkey), and so (takes effect immediately upon entry). The majority of axis configuration data used in this simulation carries the po effectiveness, meaning an NCK Reset is mandatory after each group of changes for them to take effect.

3.4.2 Recommended NC Commissioning Sequence

The recommended procedure for NC commissioning follows a defined sequence of parameterisation stages. First, axis and spindle parameter sets are configured, encompassing velocity limits, setpoint/actual-value systems, and closed-loop control data. Second, the axis configuration is established — covering coordinate systems, axis assignments, and rotary axis definitions. Third, axis-specific measurement system parameters are entered, including incremental or absolute encoder settings, position controller tuning, and axis monitoring thresholds. Fourth, spindle-specific data are parameterised, covering gear stages, velocity adaptation, and spindle positioning behaviour. Fifth, referencing of each axis and spindle is performed against the configured measuring system. For the virtual machine environment in this case study, drive-related parameters are handled by the CMVM drive replacement component, which simulates drive behaviour without requiring physical encoder or motor data. Consequently, the parameterisation was focused on axis configuration and naming for the three linear axes (MX1, MY1, MZ1), which directly governs how the NC identifies and communicates with each configured axis.

3.4.3 Accessing the Setup Operating Area in SINUMERIK Operate

All NC machine data entries in this simulation were made through the Setup operating area within SINUMERIK Operate, accessible directly from the CMVM HMI window. Within the Setup area, the Machine data submenu provides separate editors for General MD, Channel MD, and Axis MD — each corresponding to one tier of the machine data hierarchy. Navigation between axis data records within the Axis MD editor is performed using the "Axis +" and "Axis -" softkeys, which cycle through the data records associated with each configured machine axis sequentially. After completing all machine data entries, an NCK Reset was executed via the "Reset (po)" softkey to activate all changes simultaneously, rather than resetting after each individual modification.

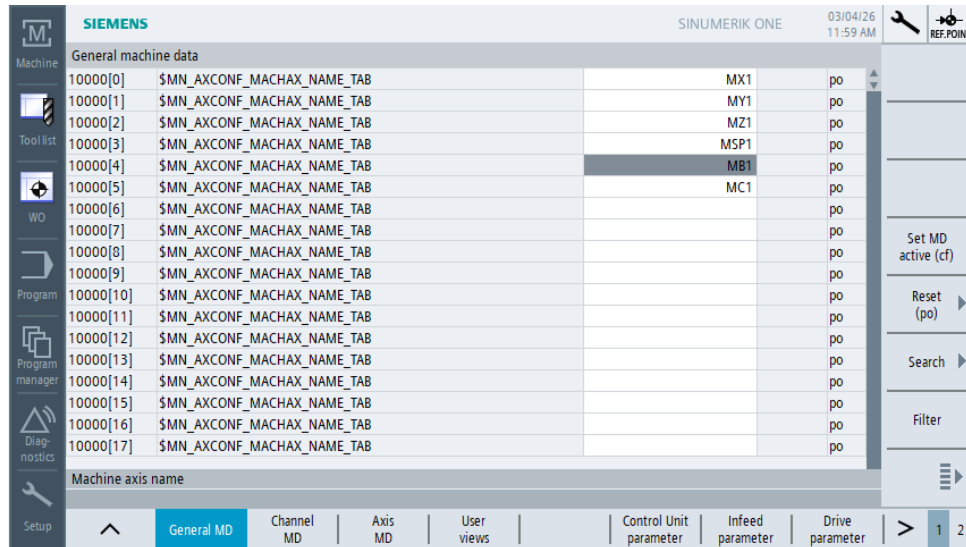


Figure 3.8: SINUMERIK Operate Setup operating area showing the Machine Data menu with General MD

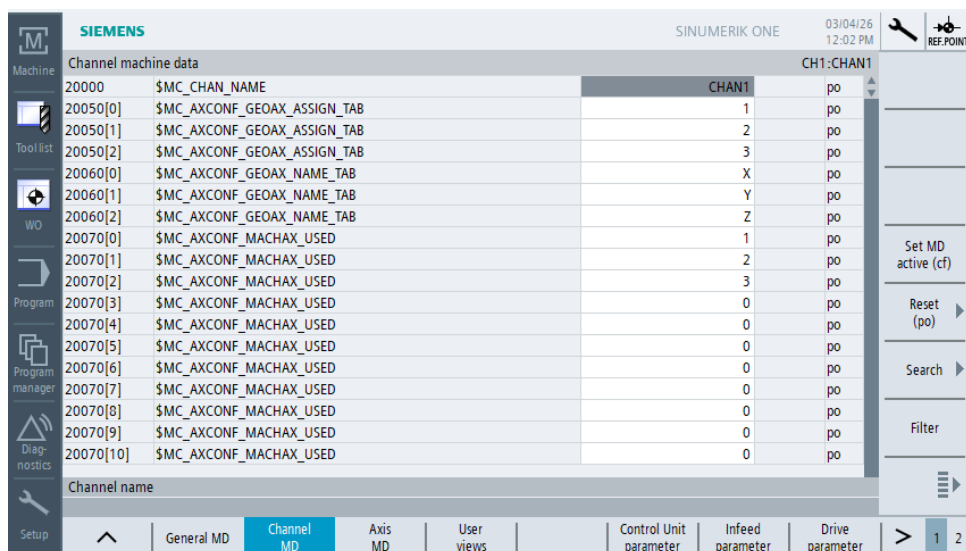


Figure 3.9: SINUMERIK Operate Setup operating area showing the Machine Data menu with Channel MD

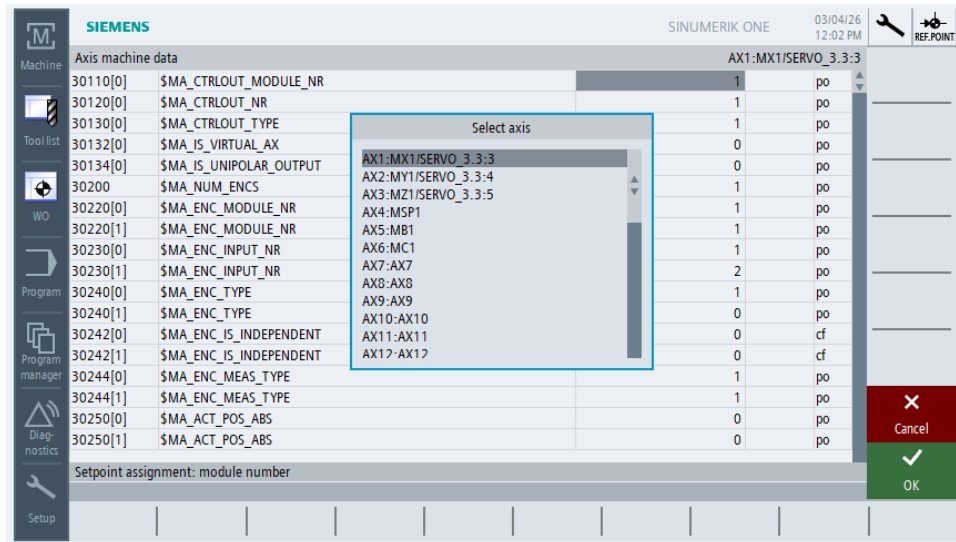


Figure 3.10: SINUMERIK Operate Setup operating area showing the Machine Data menu with Axis MD

3.4.4 Configuring Machine Axes

Axis Assignment Concept (Geometry → Channel → Machine Axes)

The SINUMERIK ONE NC operates with a three-tier axis hierarchy in which physical motion units on the machine are represented at progressively abstract levels. At the lowest level, machine axes represent the physical motion units — linear or rotary — that execute programmed traversing movements on the machine. Above them, channel axes constitute the sum of all machine, geometry, and special axes assigned to a given machining channel. Channel axes serve both the program-technological function (via geometry and special axes used in part programming) and the physical execution function (via their mapping to machine axes).

Geometry axes form the rectangular Cartesian coordinate system of the channel. A maximum of three channel axes can be designated as geometry axes, and they must be assigned to channel axes in ascending order without gaps. All channel axes that are not declared as geometry axes are classified as special axes; no geometric context is defined between special axes or between special axes and geometry axes.

The complete assignment - from geometry axes through channel axes to machine axes - is entirely governed by machine data entries. The relationship between these layers is illustrated in Figure 7-1 of the commissioning document, which maps MD entries such as MC_AXCONF_GEOAX_NAME_TAB, MC_AXCONF_CHANAX_NAME_TAB, MC_AXCONF_MACHAX_USED, and MN_AXCONF_MACHAX_NAME_TAB to their respective positions in the axis hierarchy across channels and NCU slots.

Increasing Number of Axes in Channel Machine Data (MD 20070)

By default, the SINUMERIK ONE NCU is initialised with fewer active axes than the maximum supported. To activate the four axes required for this simulation – three linear axes and one spindle – the number of machine axes assigned to channel 1 was declared through channel machine data MD 20070 (*MC_AXCONF_MACHAX_USED*). Each index of this data entry specifies which machine axis number is assigned to the corresponding channel axis slot.

The entries were made under Setup > Machine data > Channel MD for channel CH1:CHAN, as follows:

After entering all four values, an NCK Reset was executed via the "Reset (po)" softkey. Following the reset, the four axes appeared in the control display under preassigned default

Table 3.5: Channel MD 20070 configuration for CH1 (3 axes)

Channel MD	Parameter	Value	Effectiveness
20070 [0]	\$MC_AXCONF_MACHAX_USED	1	PO
20070 [1]	\$MC_AXCONF_MACHAX_USED	2	PO
20070 [2]	\$MC_AXCONF_MACHAX_USED	3	PO

names, which were subsequently replaced with the project-specific identifiers in the steps below.

Assigning Machine Axis Names (MD 10000): MX1, MY1, MZ1

Machine axis names are defined in general machine data MD 10000 (\$MN_AXCONF_MACHAX_NAME_TAB). Each index of this table corresponds to one machine axis, and the name entered is used throughout the NC for axis-specific machine data access, reference point approach from part programs (G74), test point traversing (G75), PLC-initiated axis traversal, axis-specific alarm display, actual-value system display, and the DRF handwheel function. Machine axis names must be unambiguous across the entire NC.

For this simulation, the following entries were made under Setup > Machine data > General MD:

Table 3.6: General MD 10000 configuration (3 axes)

General MD	Parameter	Value	Effectiveness
10000 [0]	\$MN_AXCONF_MACHAX_NAME_TAB	MX1	PO
10000 [1]	\$MN_AXCONF_MACHAX_NAME_TAB	MY1	PO
10000 [2]	\$MN_AXCONF_MACHAX_NAME_TAB	MZ1	PO

These names appear in the JOG/REF screen each time the machine is powered on. In part programs, axis traversal uses the channel axis names rather than these machine axis identifiers, except when traversing to the reference position via G74.

3.4.5 Assigning Channel Axis Names (MD 20080): X1, Y1, Z1

Channel axis names govern how axes are addressed within part programs and the channel's machining coordinate system. They are defined in channel machine data MD 20080 (\$MC_AXCONF_CHANAX_NAME_TAB) and must be unambiguous within the channel. These names are used for programming path axes, positioning axes, synchronized axes, spindles, and command axes in the NC part program.

The following entries were made under Setup > Machine data > Channel MD for channel CH1:CHAN:

Table 3.7: Channel MD 20080 configuration (3 axes)

Channel MD	Parameter	Value	Effectiveness
20080 [0]	\$MC_AXCONF_CHANAX_NAME_TAB	X1	PO
20080 [1]	\$MC_AXCONF_CHANAX_NAME_TAB	Y1	PO
20080 [2]	\$MC_AXCONF_CHANAX_NAME_TAB	Z1	PO

Following entry, an NCK Reset was executed to activate all machine data changes across MD 20070, MD 10000, and MD 20080 simultaneously. The Setup operating area subsequently

displayed the three axes with their newly assigned identifiers, confirming successful acceptance of the configuration.

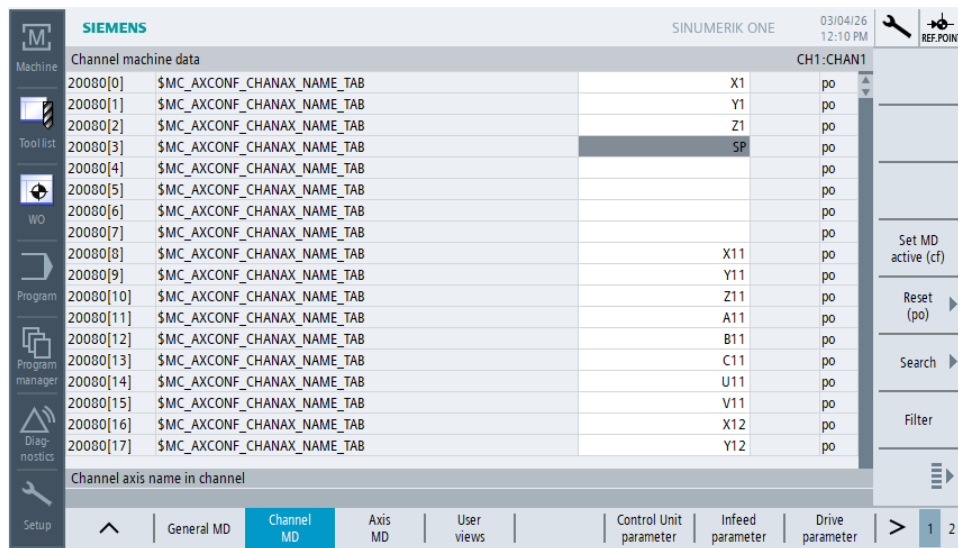


Figure 3.11: SINUMERIK Operate Setup area showing the Channel MD editor with MD 20080 entries, and the resulting axis display confirming MX1, MY1, and MZ1 as active configured axes

3.5 Engineering the HMI

3.5.1 HMI Commissioning Overview

A distinctive characteristic of the SINUMERIK ONE virtual machine environment is that the Human–Machine Interface does not require separate commissioning when the standard operator software is used. The standard HMI – SINUMERIK Operate – is automatically prepared and instantiated by CMVM V1.5.1.1 at the point of CNC software selection during machine project creation. Because CNC SW V6.25 was selected when establishing the machine project in this case study, the corresponding version of SINUMERIK Operate was automatically deployed as the HMI layer of the virtual controller without any additional configuration steps. The machine is fully operable through this standard interface from the moment the simulation is started.

This behaviour contrasts with real machine commissioning, where the HMI software must be installed on the operator panel hardware and the TCU (Thin Client Unit) connection must be configured separately. In the virtual environment, CMVM encapsulates all of this within its simulation framework, presenting SINUMERIK Operate directly within the CMVM application window.

3.5.2 Standard HMI via SINUMERIK Operate

CMVM incorporates the proven SINUMERIK Operate commissioning and operating software as an integral component of its simulation environment, supplementing it with a virtual machine control panel and the simulation control interface. Functionally, the operation within CMVM corresponds precisely to that of a real SINUMERIK ONE control equipped with a physical SINUMERIK Operate panel — all operating areas, softkey menus, machine data editors, JOG/REF screens, and alarm displays behave identically to their hardware counterparts.

In this simulation, SINUMERIK Operate served as the exclusive interface for all NC commissioning tasks described in Section 4, including navigation to the Setup operating area, access to the General MD, Channel MD, and Axis MD editors, and execution of NCK Resets. The HMI resolution was configured at the time of machine project creation to match the available monitor resolution, and the MCP483 Milling virtual control panel was embedded within the CMVM window alongside the Operate display, replicating the physical operator panel layout used on a real milling machine.

3.5.3 Accessing and Navigating the Virtual HMI in Create MyVirtual Machine

Within the CMVM application window, the SINUMERIK Operate HMI occupies the central display area, while the simulation control area — containing the Start/Stop machine button and component status indicators — is presented alongside it. The virtual MCP483 Milling panel is rendered as an interactive graphical element within the same window, with clickable buttons corresponding to axis selection, mode selection (JOG, AUTO, MDA), feedrate override, and the directional traverse keys. This layout replicates the operator experience of a physical SINUMERIK ONE milling control.

Navigation within SINUMERIK Operate follows the standard operating area structure: the main operating areas — Machine, Parameter, Program, Setup, Diagnosis, and System — are accessible via the horizontal softkey bar at the top of the display. All NC machine data entries for axis configuration were performed through the Setup operating area as described in Section 4. The Machine operating area provides the JOG and AUTO mode interfaces used during the first function test, where axis traversal via the MCP was verified after completing the NC commissioning steps.

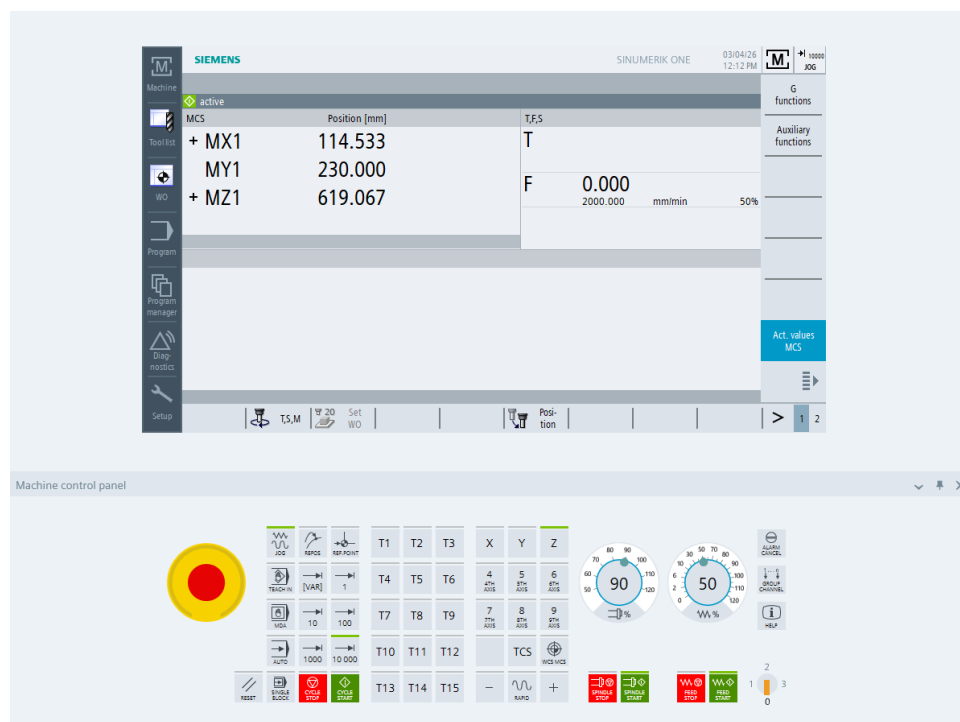


Figure 3.12: SINUMERIK Operate HMI interface running inside Create MyVirtual Machine

3.6 Loading Complete Project and Verification

3.6.1 Loading the Full TIA Portal Project (Hardware + Software) into Create MyVirtual Machine

Following successful download of the hardware configuration (Section 3.3) and completion of all user PLC program blocks, the complete TIA Portal V20 project – comprising both hardware configuration and software – was compiled and loaded into CMVM V1.5.1.1 as a unified operation. The prerequisite conditions verified before initiating this step were: CMVM was running with the machine project open and the simulation started, the hardware configuration had already been downloaded and accepted by the virtual NCU, and the machine project was in active simulation mode.

The software compilation was initiated by right-clicking the “NCU” in the project navigator and selecting “Compile > Hardware and software (changes only)”. This compiles the complete project in a single pass, validating both the hardware configuration and all software blocks – including the LBP library blocks, user-created functions AxisEnable [FC30], AxisOrganisation [FB30], BagNckAlarm [FC31], and the organisation blocks Startup [OB100], Hardware Interrupt [OB40], and Main [OB1].

The software was then transferred to the virtual machine by right-clicking the “NCU” and selecting “Load to device > Software (all)”. The subsequent download dialog sequence – Extended Load settings, target device search and selection, Load Preview, and Results dialog – followed the same procedure as the hardware configuration download described in Section 3.3. A critical distinction at this stage is that “Start PLC” must be explicitly selected in the “Results of the load procedure” dialog. This step is mandatory: the PLC will not enter RUN mode automatically upon software download unless this option is actively chosen. The dialog was then confirmed with “Finish” to complete the full project transfer.

3.6.2 Saving and Restarting the Machine Project

After the software download completed and the PLC was started, the workflow required switching to the CMVM application and performing a machine project save and restart sequence to ensure all downloaded project data became persistently active within the simulation environment.

The machine was first exited by clicking the “Exit machine” button in the CMVM simulation control area – machine projects can only be saved when the simulation control is in Stop status; saving is not possible while the simulation is running. The project was saved via “Save” or “Save as” from the CMVM menu bar, writing all current NC, HMI, PLC, and drive data to the *.vcp machine project file. Changes made to a running machine project are not saved automatically; explicit saving is required to preserve the commissioned state.

Following the save operation, the machine was restarted by clicking “Start machine” in the simulation control area. Upon restart, all downloaded project data – hardware configuration, PLC software, and the NC machine data entries configured via SINUMERIK Operate – became fully active within the virtual controller. The restart corresponds functionally to a power-on cycle of a real SINUMERIK ONE, including execution of the PLC startup OB and initialisation of the NCK with the stored machine data.

3.6.3 I/O Simulation Testing (Onboard I/O Simulation in Create MyVirtual Machine)

CMVM V1.5.1.1 provides an integrated onboard I/O simulation facility for testing the machine configuration without physical I/O hardware. I/O devices in the context of the SINUMERIK ONE encompass digital and analog input/output modules, drives, and simulator mod-

ules – all of which connect to the NCU and establish the link between the controller and the process. In the virtual machine environment, the onboard I/O simulation within CMVM substitutes for these physical devices, allowing the machine configuration to be exercised and validated entirely within the simulation.

For applications requiring more complex I/O simulation scenarios, CMVM also supports integration of external simulators via its open interface. This is particularly relevant when simulating machine-specific peripheral devices such as tool changers, workpiece clamping systems, or sensor arrays that rely on I/O signal exchange with the PLC user program. In this case study, the onboard I/O simulation was sufficient for verifying the basic axis enable logic, MCP signal routing, and emergency stop reaction programmed in Sections 3.4 and 3.5, as the machine configuration comprised three linear axes and one spindle without additional peripheral I/O requirements.

3.6.4 Verifying the Commissioning on the Virtual Machine Control Panel

The final verification of the complete commissioning workflow was performed through the first function test using the virtual MCP483 Milling control panel within CMVM. This test confirmed that the PLC–NC communication was correctly established and that all three linear axes — MX1, MY1, MZ1 — were operational.

The machine root screen was accessed in the JOG operating area of SINUMERIK Operate. The actual-value window displayed the configured linear axes MX1, MY1, and MZ1 with their current position values. Feedrate enable was activated via the virtual MCP, and the feedrate override was set to 100%. Each axis was selected individually using the axis selection keys on the MCP and traversed using the “+” and “–” directional traverse buttons, confirming that the axis enable logic in AxisOrganisation [FB30] and AxisEnable [FC30] was functioning correctly for all three linear axes.

The emergency stop function was validated by pressing the virtual E-Stop button on the MCP, which triggered alarm 3000 in SINUMERIK Operate and braked all axes, confirming correct execution of the BagNckAlarm [FC31] emergency stop network. The alarm was cleared by pressing the E-Stop button again to release it, followed by pressing the “Reset” button on the MCP to acknowledge and delete the alarm, returning the machine to a fully operational state.

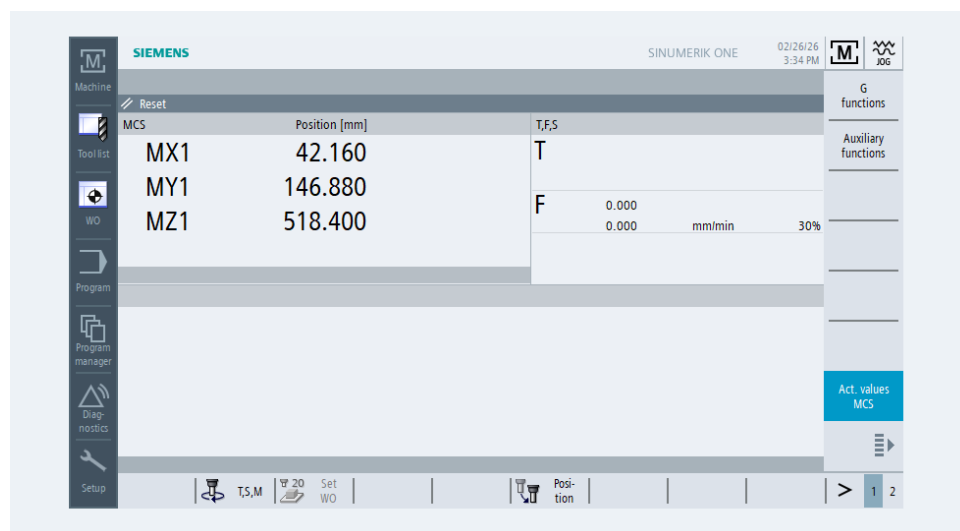


Figure 3.13: SINUMERIK Operate JOG screen within CMVM showing MX1, MY1, and MZ1 actual-value positions, confirming all three linear axes commissioned and active in the virtual controller.

3.6.5 Version Compatibility Check (CNC SW V6.25 + Toolbox V20)

Version compatibility between the CNC software and the TIA Portal project is a mandatory requirement that must be verified before any download is initiated. The versions of CNC SW and the SINUMERIK ONE STEP 7 Toolbox must be mutually compatible for the project to load and execute correctly in CMVM. For this simulation, CNC SW V6.25 was used in conjunction with SINUMERIK ONE STEP 7 Toolbox V20, with the basic PLC program at the corresponding V6.25 version level — a combination verified to be compatible prior to initiating the hardware configuration download.

The SINUMERIK ONE platform provides cross-version compatibility from CNC SW V6.13 and SINUMERIK ONE STEP 7 Toolbox V16 onwards, meaning an existing TIA Portal project configured under an older version can be loaded into a virtual NCU running a newer CNC SW version. Conversely, existing DSF backup or setup archives created with an older CNC SW version can be read into an NCU with a newer CNC SW. This forward compatibility policy ensures continuity of existing project assets when upgrading to newer software releases, and underpins the migration pathway from the virtual machine commissioning environment established in this case study to the real SINUMERIK ONE hardware platform described in the subsequent transition phase.

4 Results and Observations

4.1 Successful Virtual Commissioning Summary

The virtual machine commissioning workflow was successfully validated through the following functional tests:

- Axis JOG motion: The three linear axes MX1, MY1, MZ1 responded correctly to positive/negative jog commands from the virtual MCP483 panel, with plausible position updates and no following errors.
- Feedrate override: The feedrate override switch on the MCP correctly modulated axis traverse speeds when set to 50% and 100%.
- Emergency stop: Virtual E-Stop button triggered alarm 3000 and axis braking via Bag-NckAlarm FC31, with correct reset behaviour.

These tests confirmed correct PLC–NC interface operation, axis enable logic, and basic machine functions for the three linear axes configuration.

At the conclusion of the PLC engineering phase, the hardware configuration was accepted by the virtual NCU, the complete PLC software including LBP library blocks and user program was downloaded and the PLC entered RUN mode. The NC commissioning phase resulted in all three linear axes being correctly named, typed, and displayed in SINUMERIK Operate with their respective identifiers — MX1/X1, MY1/Y1, MZ1/Z1 — confirming that the machine data entries across General MD, Channel MD, and Axis MD were consistently applied following each NCK Reset. The HMI was operational without separate commissioning, as SINUMERIK Operate was automatically instantiated by CMVM upon CNC SW V6.25 selection. The first function test via the virtual MCP483 Milling panel confirmed that PLC–NC communication was established, all three linear axes could be traversed via the directional keys, and the emergency stop circuit functioned correctly — triggering alarm 3000 upon E-Stop actuation and clearing upon reset acknowledgement. The machine project was saved to the *.vcp file, preserving the complete commissioned state for subsequent re-use or transition to the real machine.

4.2 Issue Encountered and how it Were Resolved

The issue concerned device detection failure during the hardware configuration download. When TIA Portal V20 initiated a search for the CMVM virtual NCU via the Extended Load dialog, the target device was not detected despite the simulation being active and the IP address having been configured at X150. Investigation identified the root cause as the PLCSIM Advanced network adapter (TCP/IPv4). (TCP/IPv4) being disabled in Windows network settings. With this adapter inactive, PLCSIM Advanced could not bind correctly to the host network stack, which prevented proper communication and resulted in the expected device/-connection behavior not appearing during the setup process. The solution was to open the Windows network adapter settings, locate the PLCSIM Advanced adapter, and enable (TCP/IPv4). (TCP/IPv4). After activating the adapter, the subsequent attempt worked as expected.

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